

Curriculum Vitae



General Information

First and Last Name
Email
Website · Linkedin · Github

Dr. Mikel Sagardia
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Work Experience

Period
Position Held
Employer
Address of Employer
Main Activities and Responsibilities

2023.05.01 – present
Senior Data Scientist and Machine Learning Engineer
Deutsche Telekom
Friedrich-Ebert-Allee 140, 53113 Bonn (Germany) · Remote from Spain
– Design and development of object detection applications
– Research and development of vectorization algorithms for images and text
– Implementation of data clustering applications
– Development and deployment of LLM and RAG-based applications

Period
Position Held
Employer
Main Activities and Responsibilities

2022.04.01 – 2023.04.30 (1 year and 1 month)
Computer Vision and Machine Learning Engineer
Self Employed
– Built machine vision and machine learning applications as a contractor
– Supported and trained companies in the field of deep learning

Period
Position Held
Employer
Address of Employer
Main Activities and Responsibilities

2020.05.01 – 2021.12.31 (1 year and 8 months)
Senior Researcher
Vicomtech, Fundación
Mikeletegi 57, 20009 Donostia-San Sebastián (Spain)
– Coordinated and executed European and national projects
– Acquired new projects on computer vision, machine learning and robotics
– Supervised Ph.D. students

Period
Position Held
Employer
Address of Employer
Main Activities and Responsibilities

2019.03.01 – 2020.03.31 (1 year and 1 month)
Project Engineer in Machine Vision
MVTec Software GmbH
Arnulfstr. 205, 80634 Munich (Germany)
– Managed and developed machine vision projects for key industry partners
– Trained customers and distributors on 3D vision and robotics
– Presented the Anomaly Detection feature at the MVTec Innovation Day 2020
– Provided second level technical support in a broad spectrum of machine vision topics

Period
Position Held
Employer
Address of Employer

2008.04.01 – 2018.12.31 (10 years and 9 months)
Research Engineer in Robotics
German Aerospace Center (DLR), Institute of Robotics and Mechatronics
Muenchner Str. 20, 82234 Wessling (Germany)

Main Activities and Responsibilities

Achievements and Awards

- Led a 5-year project that built a VR framework for satellite on-orbit servicing
- Researched into collision and force computation methods for haptic rendering
- Coordinated and executed cooperation projects for industry partners (VW, Bosch)
- Published of 22+ papers at IEEE T. on Haptics / ICRA / VR, ACM VRST
- Presented robotic applications on trade fairs and congresses (Automatica, VW, etc.)
- Supervised 20+ students during their master's theses and internships
- Best Video Award Finalist at the IEEE ICRA, 2014
- Best Paper Award at the 5th International Conference VAMR, 2013
- Short Paper Honorable Mention at the IEEE VR, 2012

Education and Training

Period
Qualification
Selected Courses

2017.08 – present

MOOC Certificates · github.com/mxagar/course_compilation

- Deep Learning Nanodegree (150+ hours) · Udacity
- Computer Vision Nanodegree (120+ hours) · Udacity
- Data Science Nanodegree (150+ hours) · Udacity
- Machine Learning DevOps Engineer Nanodegree (150+ hours) · Udacity
- IBM Machine Learning Professional Certificate (85+ hours) · Coursera / IBM

Period
Qualification
University
Dissertation Title

2013.12 – 2018.11

Ph.D. in Electrical and Information Engineering (Dr.-Ing.), 2019

Technische Universität München, TUM (Germany)

Virtual Manipulations with Force Feedback in Complex Interaction Scenarios

Period
Qualification
Universities

2002.09 – 2008.02

M.Sc. in Mechanical Engineering (Ingeniero Industrial), 2008

Universidad de Navarra, TECNUN (Spain) and
Technische Universität München, TUM (Germany)

Achievements, Awards and Scholarships

- Second Best Diploma Thesis Prize · Association of Engineers of Gipuzkoa
- Erasmus and Leonardo scholarships, exchange student at TUM
- High School Excellence Award for university access · Basque Government

Technical Skills

Programming Languages
Machine/Computer Vision
Machine Learning
DevOps, MLOps
Web Development
Cloud & Big Data
Software Architecture

Python, C/C++, SQL, Bash, R

OpenCV, Halcon, Open3D, PCL

Pytorch & Lightning, Tensorflow & Keras, Scikit-Learn, Pandas, Matplotlib, Numpy, SciPy, NLTK, spaCy, HuggingFace, LangChain

Docker & Kubernetes, Git & CI/CD, MLflow, W. & Biases, DVC, Pytest

FastAPI, Flask, Streamlit, Gradio, HTML & CSS

AWS, Azure, PySpark, ElasticSearch

Object Oriented Programming, Design Patterns, Data Structures

Languages

Mother Tongues
Advanced Level (C1/2)

Spanish, Basque (*Euskararen Gaitasun Agiria* · EGA)

German (*Goethe-Zertifikat C1*), **English** (Certificate in Advanced English · CAE)

Additional Information

Publications
Selected Projects

<https://mikelsagardia.io/publications/>

https://github.com/mxagar/project_compilation